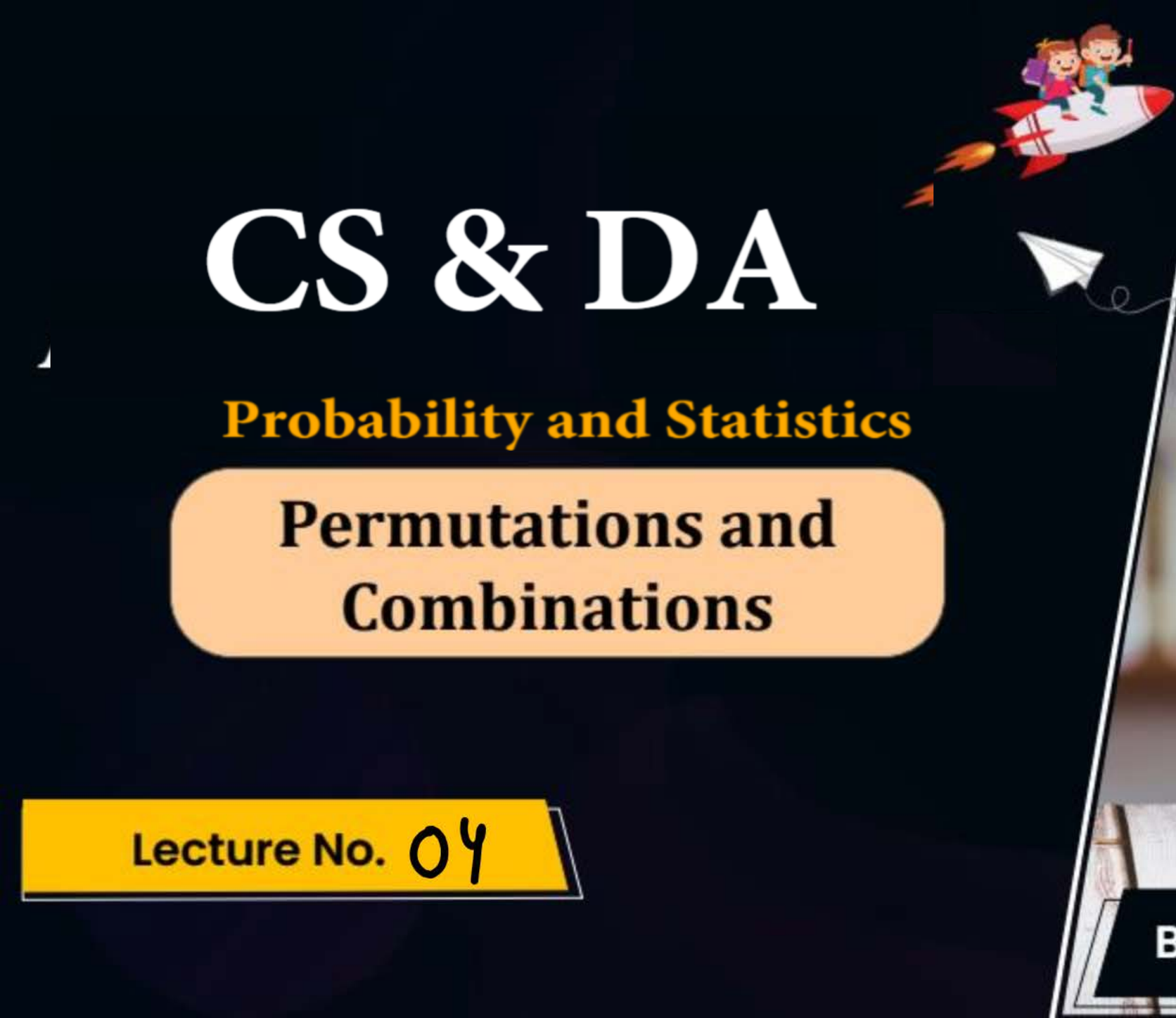
CS & DA

Probability and Statistics

Permutations and Combinations

Lecture No. 04



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

1. Counting Principle → Addition P, Multiplication P
2. Permutation (selection & Arrangement)
3. Combination (only selection)
4. Concept of At least & At Most
5. De-Arrangements
6. Perm of Alike items.

Topics to be Covered



Topic

BASICS of P & C (Part 4)

RECAP

(✓) Number of linear Arrangements of n different things $= n!$

(✓) Number linear Arrangements of n things in which p are alike, q are alike, r things are alike & Rest are different =

$$= \frac{{}^{n}P_{n}}{p!q!r!} = \frac{n!}{p!q!r!}$$

where $p+q+r+\underset{(?)}{\text{Rest}} = \text{Total}(n)$

(✓) No. of De-Arrangements of n things $= n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\cdots\cdots+(-1)^{n}\frac{1}{n!}\right]$

Eg: $n=6$ persons Then Total arrangements $= 6! = 720$

& All correct arrangements $= 1$

Dearrangements (All wrong Arrangements) $= 6!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right] = 265$

Analysis:-

(No wrong) or (2 wrong) or (3 wrong) or (4 wrong) or (5 wrong) or (All wrong) = Total Cases

(= All Correct)

At least 2 wrong

At most 3 wrong

At least 4 wrong

$${}^6C_0 \times 1 + \left({}^6C_2 \times 1\right) + \left({}^6C_3 \times 2\right) + \left({}^6C_4 \times 9\right) + \left({}^6C_5 \times 44\right) + \left({}^6C_6 \times 265\right) = 720$$

$$1 + (15 \times 1) + (20 \times 2) + (15 \times 9) + (6 \times 44) + (1 \times 265) = 720$$

(1) No. of Arrangements in which at least 2 persons will be seated on wrong chair $= ? = 720 - 1 = 719$

(2) " " " " exactly 2 persons " " " " " $= ? = {}^6C_2 + 1 = 15$

(3) " " " " At least 4 persons " " " " " $= ?$

$= {}^6C_4 \times 9 + {}^6C_5 \times 44 + {}^6C_6 \times 265 = ?$

(4) " " " " exactly 4 Persons will be seated on wrong chair $= {}^6C_4 \times 9$

(5) " " " " At Most 3 Persons " " " " " $= ?$

$= ({}^6C_0 \times 1) + ({}^6C_2 \times 1) + ({}^6C_3 \times 2)$

Note ① At Most → Include None also. (in general)

② At least → Go upto Last point

③ (0 App) or (1 App) or (2 App) or (3 App) or (4 App) or (5 App) = Total Possibilities

(0 App) or (1 App) or (2 App) → At most 2

(3 App) or (4 App) or (5 App) → At least 3

(0 App) or (1 App) or (2 App) or (3 App) → At Most 3

(4 App) or (5 App) → At least 4

eg (At Most 2) + (At least 3) = Total (5 App)

g (At Most 3) + (At least 4) = Total (5 App)

g (At Most 1) + (At least 2) = Total (5 App)

(At Most r) + (At least r+1) = Total (n)

Some Important Points

1. Always together (Not separated) → Assume them as one unit within Bracket
2. All Never together (All Do not Come together) → Total − Always together
3. No two Girls are together → first arrange Boys
4. Alternately (Linear Case) → Two Cases will arise
5. Alternately (Circular Case) → only one Case will arise
6. Particular / (fix) → No Need to Select & No Need to arrange
7. Alike → use the Result $\frac{n!}{p!q!r!}$

2B & 2G :-

All G are together :-

$(B_1\ B_2\ G_1\ G_2), (B_2\ B_1\ G_1\ G_2)$

$(B_1\ B_2\ G_2\ G_1) \rightarrow (B_2\ B_1\ G_2\ G_1)$

$(B_1\ G_1\ G_2\ B_2) \rightarrow (B_2\ G_1\ G_2\ B_1)$

$(B_1\ G_2\ G_1\ B_2) \rightarrow (B_2\ G_2\ G_1\ B_1)$

$(G_1\ G_2\ B_1\ B_2) \rightarrow (G_1\ G_2\ B_2\ B_1)$

$(G_2\ G_1\ B_1\ B_2) \rightarrow (G_2\ G_1\ B_2\ B_1) \Rightarrow$ Total Case = 12 Cases

M-II / Trick

$$\underset{1}{B_1}\ \underset{2}{(G_1\ G_2)}\ \underset{3}{B_2} = 3! \times 2! = 12$$

Q. FQ 1 4 Boys and 4 Girls are to be seated in a Row then find the Total number of arrangements

if ① there is No Restriction = ? = $^8P_8 = 8!$ (Man Ans)
(RNA)

② All Boys are together & All Girls are together = ? = $2! \times 4! \times 4!$ = 1152 ways
$(B_1 B_2 B_3 B_4), (G_1 G_2 G_3 G_4)$
(Persons) (Boys) (Girls)

③ All Girls are together = ? = $5! \times 4!$ = 2880
(Persons) (Girls)

$\underline{B_1}, \underline{B_2}, \underline{(G_1 G_2 G_3 G_4)}, \underline{B_3}, \underline{B_4}$
1st 2nd 3rd 4th 5th Person

(4) All Girls are Never together = ? = Total − Girl are together

$= 8! - (5! \times 4!) = 37440$

(5) No Two Girls are together = ? = First arrange Boys $= 4! \times ({}^{5}C_4 \times 4!)$

$\checkmark B_1 \checkmark B_2 \checkmark B_3 \checkmark B_4 \checkmark$

$= 4! \times {}^{5}P_4 = 2880$

(Boys) (Girls)

(6) Boys and Girls are Seated alternately = ?

Case I: (B G B G B G B G) → $4! \times 4!$ (1st, 3rd, 5th, 7th)

or

Case II: (G B G B G B G B) → $4! \times 4!$ (2nd, 4th, 6th, 8th)

$A_6 = 2 \times 4! \times 4! = 1152$

✱ Part (5) is a particular case of part (4)

Analysis of part (5) → No two G are together = ? = 2880

Case I: (B G B G B G B G) → $4! \times 4! = 576$ Cases

or Case II: (G B G B G B G B) → $4! \times 4! = 576$ Cases

or Case III: (G BB G B G B G) → ? = 576 Cases

or Case IV: (G B G BB G B G) → ? = 576 "

or Case V: (G B G B G BB G) → ? = 576 "

+

= 2880 Cases

✱ Part (6) is a particular case of Part (5)

(7) out of 4G if there are two sisters then find the No. of arrangements if they are not separated = ?

eg G G (S1 S2) B B B B

1 2 3rd 4 5 6 7th

$= 7! \times 2!$

(Persons)(Sisters)

(8) Two sisters are separated by exactly one person = ?

eg P P (S1 P S2) P P P

1 2 3rd 4 5 6th

$= ({}^{6}C_1 \times 1) \times 6! \times 2!$

(P) (Sist)

(9) Two sisters are separated by exactly one Boy = ?

eg B G (S1 B S2) B B G

$= ({}^{4}C_1 \times 1) \times 6! \times 2!$

(10) " " " " " exactly one Girl = ? $= ({}^{2}C_1 \times 1) \times 6! \times 2!$

(11) there is a particular person b/n two sisters = ? = $6! \times 2!$

(Per) (Sist)

P P (S1 (P) S2) P P P

1 2 3 4 5 6 th

(12) Two sisters wants to be seated at the adjacent sides of their Brother = ?

B B (SE (B) SY) G B G = $6! \times 2!$

No Need to select Bro bcoz Brother is Particular Person & he knows it.

(13) Elder & Younger sister has to be seated at 1st & last position Resp = ?

eg { (SE) (G G B B B B) (SY) } = $6! = 720$

(14) Two sisters wants to be seated at extreme positions = ?

eg { S_1 (G G B B B B) S_2 } $= \underset{(P)}{6!} \times \underset{(S)}{2!} = 720 + 720 = ?$

(15) Two sisters wants to be seated at extreme positions But together = ?

Case I $(S_1\ S_2)$ B B B G G B $= 6! \times 2!$

1 2 3 4 5 6th

or

Case II B B G B B G $(S_1\ S_2)$ $= 6! \times 2!$

1 2 3 4 5 6th

Ans $= 2 \times 6! \times 2!$

HW8 16 persons are to be seated along a long Table with 8 chairs on each side. Four particular Persons wants to be seated on one particular side while 2 persons on the other side. Then find the possible Number of Arrangements ?



Ans $(94080 \times 10!)$

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THANK - YOU